

Home Lab Safety Agreement

Sign once, at or before the first practice. Home-lab homework cannot be assigned to a student until this is signed and returned to the coach.

What the home-lab program is

Crime Busters is a laboratory event. About half the competition score comes from identifying unknown powders, liquids, and metals by running quick chemical and physical tests. Kids get good at this by *doing it*, many times. To make that practice possible at home between our weekly meetings, the coach provides each student a small **home practice kit**.

What is in the kit (provided by the coach)

- A set of small, labeled bags of **food-grade powders**: baking soda, corn starch, table salt, granulated sugar, powdered (confectioner's) sugar, Epsom salt, cream of tartar, citric acid, chalk / calcium carbonate, clean play sand.
- A dropper bottle of **tincture of iodine** (the same strength sold at any pharmacy as a first-aid antiseptic).
- A small bottle of **red-cabbage pH indicator** (a purple liquid the coach makes by boiling red cabbage; it is just cabbage and water).
- A few sheets of coffee-filter paper and a set of labeled "unknown" bags for practice sessions.

What the family provides (ordinary kitchen items)

- White vinegar
- Water
- A refrigerator magnet or other small magnet
- Paper towels, a few clear cups or a muffin tin, toothpicks, a spoon

What is NOT in the kit and never goes home

- **No strong acids or bases.** The 1 M hydrochloric acid used at competition is practiced **only at in-person practice**,

with the coach present and everyone in goggles. Do not try to substitute a stronger acid at home. Vinegar is the home stand-in and it is enough for practice.

- No open flames. The burn tests for fibers and plastics are demonstrated only at practice. Do not do burn tests at home.
- No heating, no microwaving of any sample.

Rules for every home lab (student and adult read together)

1. **An adult is in the room the entire time.** This is not optional. The home lab is a supervised activity, like using the stove.
2. **Safety goggles or well-fitting glasses on** before opening any bag or bottle.
3. **Never taste anything.** Even though the powders are food-grade, in the lab they are "unknowns" and the rule is no tasting, ever. This is a habit that matters at competition, where samples are *not* food-grade.
4. **Smell by wafting** — hand over the cup, waft the air toward your nose. Never put your nose to a sample.
5. **Work in a ventilated spot** — a kitchen counter with a window cracked or the fan on. Iodine and vinegar both have strong smells.
6. **Iodine stains** skin, clothes, and counters brown. Work on a paper towel or a tray, wear old clothes, wash hands after.
7. **Small amounts.** A pinch of powder and a few drops of liquid is plenty.
8. **Clean up completely.** Rinse everything down the sink with lots of water, wipe the surface, wash hands. Practicing cleanup matters — an unclean station loses points at competition.
9. **Keep the kit up high / out of reach** of younger siblings and pets between practice sessions.
10. **If anything gets in eyes or mouth:** rinse with lots of water and call the coach and, if you're worried at all, Poison Control at **1-800-222-1222**.

Agreement

I have read the rules above with my child. I understand that:

- home-lab homework is a **supervised** activity and an adult will be present for all of it,
- the kit contains only food-grade powders and mild household-strength iodine and a cabbage-water indicator,
- no strong chemicals, heat, or flames are involved,
- and I am responsible for storing the kit safely between practices.

Student name: _____

Parent/guardian name: _____

Parent/guardian signature: _____ Date:

Coach contact: _____